

## Exercise Sheet 3

### Smoothed Analysis of the Simplex Method

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This sheet restates the four exercises of Lecture 3. We recall the objects they concern. For a flow network  $(G, u, s, t)$  with arc capacities  $u > \mathbf{0}$  and costs  $c \in (0, 1)^E$  (where  $E$  contains at most one of each opposite pair  $ij, ji$ ),  $P$  denotes the polytope of feasible  $s$ - $t$  flows,  $\text{val}(f)$  the value of  $f$ , and  $\Phi(v) := \min\{c^\top f : f \in P, \text{val}(f) = v\}$  the (convex, piecewise linear) cost curve. The *residual network*  $N[f]$  has arcs in  $E \cup \overleftarrow{E}$ , where  $\overleftarrow{E} := \{ji : ij \in E\}$  and  $c_{ji} := -c_{ij}$ ; it contains  $a \in E$  if  $f_a < u_a$  and  $\overleftarrow{a} = ji$  if  $f_{ij} > 0$ . The *successive shortest path* (SSP) algorithm starts at  $f = \mathbf{0}$  and, while  $N[f]$  contains an  $s$ - $t$  path, augments  $f$  as much as possible along a minimum-cost  $s$ - $t$  path of  $N[f]$ ; the *uniqueness assumption* of the lecture states that minimum-cost flows of each value and minimum-cost residual paths are unique. A path or cycle may traverse an arc *forwards* or *backwards*; its signed indicator  $\chi$  records  $+1$  and  $-1$  accordingly. The lecture's *shadow path* for a pair  $(c, d)$  starts at the  $c$ -minimizer and follows, at each vertex  $v$ , the edge solving the *tangent cone program*  $\min\{c^\top y : y \in T_P(v), d^\top y = 1\}$ , where  $T_P(v)$  is the cone of feasible directions; SSP corresponds to  $d = \text{val}$ . For a linear subspace  $W \subseteq \mathbb{R}^n$ , a *circuit* is a support-minimal nonzero  $g \in W$ ;  $g$  *conforms* to  $w$  if  $\text{supp}(g) \subseteq \text{supp}(w)$  and  $g_j w_j \geq 0$  for all  $j$ ; and the *conformal decomposition lemma* states that every  $w \in W \setminus \{\mathbf{0}\}$  is a sum of at most  $\dim(W)$  circuits of  $W$  conforming to  $w$ , cancellation-free. The *circuit imbalance*  $\kappa(W) := \max |g_j / g_i|$  over circuits  $g$  and  $i, j \in \text{supp}(g)$ , with  $\kappa(\mathbf{A}) := \kappa(\ker(\mathbf{A}))$ ; as shown in the lecture via Cramer's rule,

$$\kappa(\mathbf{A}) = \max \left\{ \frac{|\det(\mathbf{A}_{B-j+i})|}{|\det(\mathbf{A}_B)|} : B \text{ basis, } i \notin B, j \in B \right\}, \quad B - j + i := (B \setminus \{j\}) \cup \{i\},$$

the maximum ratio between the determinants of two bases differing in one column.

**Exercise 1** (SSP implements the shadow vertex method). Let  $f$  be a vertex of  $P$  on the cost curve  $\Phi$  (a minimum-cost flow of its value), not of maximum value. By the variable-bound description of the tangent cone, the tangent cone program at  $f$  for the pair  $(c, \text{val})$  reads

$$\begin{aligned} \min \quad & c^\top y \\ \text{s.t.} \quad & (\mathbf{A}_G y)_v = 0 \quad \forall v \in V \setminus \{s, t\}, \quad \text{val}(y) = 1, \\ & y_a \geq 0 \quad \forall a \text{ with } f_a = 0, \quad y_a \leq 0 \quad \forall a \text{ with } f_a = u_a. \end{aligned}$$

- (a) Show first that a flow is a minimum-cost flow of its value if and only if its residual network contains no negative-cost cycle. (Hint: Decompose  $f' - f$  into conformal cycles for a competitor  $f'$ .)
- (b) Show, using the conforming-cycle part of the flow decomposition theorem (applied after appending an artificial arc  $ts$  of flow  $\text{val}(y)$ ), that the paths and cycles in the decomposition of any feasible  $y$  travel only along residual arcs of  $N[f]$ .
- (c) Show that the decomposition of a feasible  $y$  consists of  $s$ - $t$  paths of  $N[f]$  of total weight 1 together with residual cycles.

- (d) Conclude with (a) that the optimal value of the program is the minimum cost of an  $s$ - $t$  path in  $N[f]$ , attained at  $y = \chi^Q$  for the cheapest path  $Q$ : the shadow vertex pivot at  $f$  is a maximal augmentation along the minimum-cost  $s$ - $t$  residual path — an SSP iteration. Deduce the lecture's theorem: the SSP iterates are exactly the breakpoints of  $\Phi$ , in order, and the augmenting path costs are its slopes, with  $0 < s_1 < s_2 < \dots < n - 1$ .

**Exercise 2** (Source and sink arcs are charged only once). The charging analysis of the smoothed SSP theorem charges each iteration to an *arc side* in  $E \cup \overleftarrow{E}$  — the residual arc, used by the augmenting path, whose associated bound ( $f_{ij} \geq 0$  for  $ij \in E$ ,  $f_{ij} \leq u_{ij}$  for  $ji \in \overleftarrow{E}$ ) was tight before the iteration.

- (a) Show that over the run of SSP, the flow on every arc leaving  $s$ , and on every arc entering  $t$ , is monotonically nondecreasing.
- (b) Conclude that each side of such an arc is charged at most once over the entire run — the forward side at most once, the reversed side never — so these sides need no randomness at all. Deduce an alternate proof that the deterministic zero-cost auxiliary arcs of the reduction from minimum-cost flow are harmless: with only the original arcs smoothed, the expected number of iterations is still  $O(mn\phi)$ .

**Exercise 3** (Flow decomposition is conformal decomposition). Deduce the flow decomposition theorem from the conformal decomposition lemma applied to the space of circulations  $W = \ker(\mathbf{A}_G)$ , where  $\mathbf{A}_G \in \mathbb{R}^{V \times E}$  is the node-arc incidence matrix of  $G$ .

- (a) Show that the circuits of  $\mathbf{A}_G$  are precisely the nonzero scalar multiples  $t\chi^C$  of signed indicator vectors of simple cycles  $C$ .
- (b) Conclude the conforming-cycle statement: every circulation  $w$  is a positive combination of at most  $m$  simple cycles conforming to  $w$  — traversing each arc  $a$  forwards where  $w_a > 0$  and backwards where  $w_a < 0$ .
- (c) Derive the path/cycle form for a flow  $f \in P$  with  $\text{val}(f) = \nu \geq 0$ : append an artificial arc  $ts$  with  $\tilde{f}_{ts} := \nu$ , check that  $\tilde{f}$  is a circulation, apply (b), and classify the cycles by whether they use  $ts$ .

**Exercise 4** (Circuit imbalance is self-dual). Prove that for every linear subspace  $W \subseteq \mathbb{R}^n$ ,

$$\kappa(W) = \kappa(W^\perp).$$

(Hint: Choose coordinates so that  $W = \ker(\mathbf{A})$  for  $\mathbf{A} = [\mathbf{I}_m \mid \mathbf{M}]$ . Give a representation of  $W^\perp$  as the kernel of a matrix depending on  $\mathbf{M}$ , and compare its subdeterminants with those of  $\mathbf{A}$ .)